KNN and K-means from Scratch, and EM for a Coin Mixture

Machine Learning, Homework 7

Arshia Goshtasbi • MSc Electrical Engineering (Control), Iran University of Science and Technology •
Spring 1404 (2025)

Notebooks.

- `knn-from-scratch.ipynb`: the same employee-attrition data as HW3. I removed constant columns, one-hot encoded the categorical ones, made an 80/20 split and implemented KNN (Euclidean distance, majority vote) from scratch. With $K = 5$ on unscaled features, my implementation and scikit-learn's `KNeighborsClassifier` give identical results: 0.821 training and 0.762 test accuracy.
- `kmeans-from-scratch.ipynb`: K-means from scratch on 4000 random rows of California Housing, compared with scikit-learn's `KMeans` using the elbow and silhouette methods (Figure 1). Both implementations give the same curves; the silhouette score is highest at $k = 2$. I prefer the silhouette method because it gives a clear maximum, while the elbow is a judgement call.

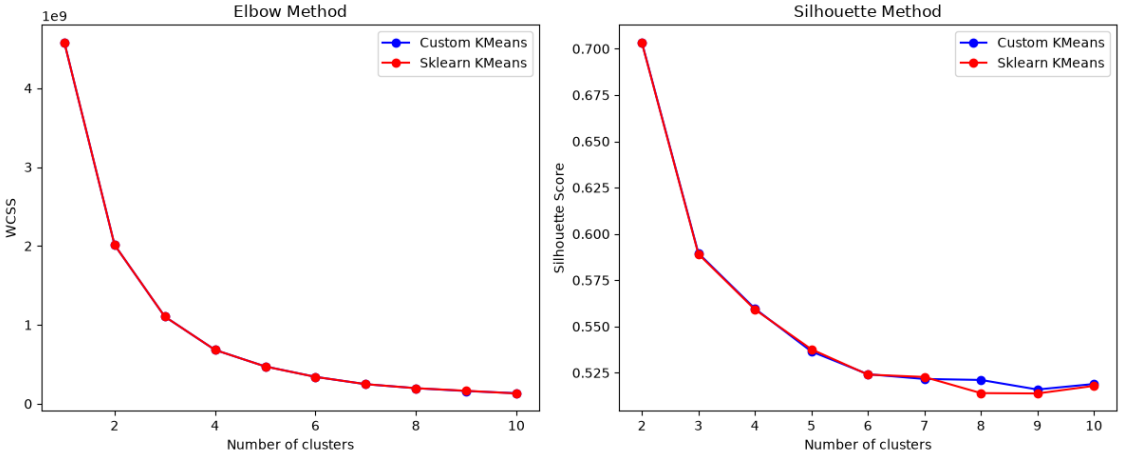


Figure 1: Elbow (WCSS) and silhouette scores for my K-means and scikit-learn's.

Written part.

KNN on a small 2D set. I found the K with the lowest training error and the lowest leave-one-out error, and drew the 1-NN decision boundary. A very small K follows noise and outliers. A very large K over-smooths and predicts the majority class everywhere.

K-means. Each step does not increase J and there are at most k^n assignments γ, so the algorithm stops in finitely many steps. The total scatter splits as $T(X) = \sum_j \frac{n_j}{n} W_j(X) + B(X)$, so minimising the weighted within-cluster scatter is the same as maximising the between-cluster scatter B. The minimum of J never increases with k (it reaches 0 at $k = n$), so k cannot be chosen by minimising J.

EM for two biased coins. With hidden coin choice Z_i and $\theta = (\pi, p_r, p_b)$: $p(x, z; \theta) = \left[\pi p_r^x (1 - p_r)^{1-x}\right]^z \left[(1-\pi) p_b^x (1 - p_b)^{1-x}\right]^{1-z}$. If z_i were known, the ML estimates are $\hat{\pi} = \frac{1}{m}\sum z_i$, $\hat{p}_r = \sum z_i x_i / \sum z_i$ and $\hat{p}_b = \sum (1 - z_i) x_i / \sum (1 - z_i)$. EM replaces z_i by the posterior $\gamma_i = P(Z_i{=}1 \mid x_i; \theta^\alpha)$ in the E-step and uses the same formulas with γ_i in the M-step.